

CAREER: Heterogeneous Hardware for Secure Autonomy at the Edge

Md Tanvir Arafin
George Mason University

Computing hardware for modern-day robotics is experiencing a paradigm shift. As compute demand increases, hardware targeted for robotic and autonomous systems shifts from a monolithic CPU-based design to a heterogeneous systems-on-chip (SoCs) architecture comprising a CPU, GPUs, neural processing units (NPU), FPGAs, and custom ASICs. This impressive increase in capabilities has enabled the deployment of critical, compute-intensive workloads on edge robotics hardware, allowing them to run billion-parameter foundation models and perform simultaneous localization and mapping (SLAM) in tandem for real-time applications.

Interestingly, although there has been tremendous progress in edge intelligence for robotics, the secure deployment of these primitives is still lagging. Common security practices in this domain still rely on classical cryptographic schemes and ad hoc notions of security, based on (outdated) performance assumptions about limited hardware resources. On the other hand, the advent of post-quantum cryptography (PQC), zero-knowledge proofs (ZKPs), and fully-homomorphic encryption (FHE) standards has given rise to an interesting research problem: how to deploy modern cryptographic primitives efficiently to protect the next generation of intelligent autonomous and robotic systems at the edge? To address this, hardware-software co-design and optimization of post-quantum-secure cryptographic primitives must be investigated thoroughly on heterogeneous SoCs. This task is broad in scope; therefore, for this project, we define three key thrusts that align with the CAREER program and provide foundational advancement towards the next generation of hardware-assisted secure autonomy.

First, we will explore the hardware requirements for zero-knowledge proof protocols that enable verifiable inference, location services, and SLAM execution. The goal of this task is to design a fully streaming, memory-first prover hardware that integrates fast Reed-Solomon IOP of proximity (FRI), hashing, and trace generation into a single pipeline on GPUs and FPGAs. In the second thrust, we will leverage this prover hardware and integrate it with the Robot Operating System (ROS) to build an end-to-end testbed for verifiable computation. This overall testbed will provide us with additional opportunities for hardware-software optimization. Finally, we will explore the compiler primitives to develop solutions that will integrate the hardware-software co-design formulation with standard compilation primitives. The products of this project will be integrated into undergraduate and graduate courses at Mason and existing NSF-supported research and training programs led by the PI for broader impact.